Q1. JEE Main 2026 (21 January Shift 2)

If P is a point on the circle $x^2+y^2=4$, Q is a point on the straight line $5x+y+2=0$ and $x-y+1=0$ is the perpendicular bisector of PQ, then 13 times the sum of abscissa of all such points P is ____.

Q2. JEE Main 2026 (28 January Shift 2)

Let the circle $x^2+y^2=4$ intersect x-axis at the points $\mathrm{A}(\mathrm{a},0), \mathrm{a}>0$ and $\mathrm{B}(\mathrm{b},0)$. Let $\mathrm{P}(2\cos\alpha, 2\sin\alpha)$, $0<\alpha<\frac{\pi}{2}$ and $\mathrm{Q}(2\cos\beta, 2\sin\beta)$ be two points such that $(\alpha-\beta)=\frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

(1) $x^2+y^2-4y-4=0$
(2) $x^2+y^2-4x-4y=0$
(3) $x^2+y^2-4x-4=0$
(4) $x^2+y^2-4x-4y-4=0$

Q3. JEE Main 2026 (22 January Shift 1)

Let the set of all values of r, for which the circles $(x+1)^2+(y+4)^2=r^2$ and $x^2+y^2-4x-2y-4=0$ intersect at two distinct points be the interval (α,β). Then $\alpha\beta$ is equal to

(1) 21 (2) 24 (3) 20 (4) 25

Q4. JEE Main 2026 (23 January Shift 2)

If the points of intersection of the ellipses $x^2+2y^2-6x-12y+23=0$ and $4x^2+2y^2-20x-12y+35=0$ lie on a circle of radius r and centre (a,b), then the value of $ab+18r^2$ is

(1) 52 (2) 53 (3) 51 (4) 55

Q5. JEE Main 2026 (21 January Shift 1)

Let $\vec{\mathrm{c}}$ and $\vec{\mathrm{d}}$ be vectors such that $|\vec{\mathrm{c}}+\vec{\mathrm{d}}|=\sqrt{29}$ and $\vec{\mathrm{c}}\times(2\hat{i}+3\hat{j}+4\hat{k})=(2\hat{i}+3\hat{j}+4\hat{k})\times\vec{\mathrm{d}}$. If $\lambda_1,\lambda_2\,(\lambda_1>\lambda_2)$ are the possible values of $(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})$, then the equation $\mathrm{K}^2x^2+\left(\mathrm{K}^2-5\,\mathrm{K}+\lambda_1\right)xy+\left(3\,\mathrm{K}+\frac{\lambda_2}{2}\right)y^2-8x+12y+\lambda_2=0$ represents a circle, for K equal to :

(1) 1 (2) 4 (3) -1 (4) 2

Q6. JEE Main 2026 (21 January Shift 1)

Let PQ and MN be two straight lines touching the circle $x^2+y^2-4x-6y-3=0$ at the points A and B respectively. Let O be the centre of the circle and $\angle \mathrm{AOB}=\pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :

(1) $x^2+y^2-12x-18y-25=0$
(2) $3\left(x^2+y^2\right)-18x-12y+25=0$
(3) $3\left(x^2+y^2\right)-12x-18y-25=0$
(4) $x^2+y^2-18x-12y-25=0$

Q7. JEE Main 2026 (28 January Shift 1)

Let $y=x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point $(2,3)$ and is farthest from the center of C_2, is $x+ay+b=0$, then $a-b$ is equal to

(1) -2 (2) 10 (3) -6 (4) 6

Q8. JEE Main 2026 (24 January Shift 1)

Let $\mathrm{S}=\left\{z \in \mathbb{C}:\left|\frac{z-6i}{z-2i}\right|=1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right|=\frac{3}{5}\right\}$.
Then $\sum_{z \in \mathrm{S}}|z|^2$ is equal to

(1) 385 (2) 398 (3) 413 (4) 423

Q9. JEE Main 2026 (24 January Shift 1)

Let a circle of radius 4 pass through the origin O, the points $\mathrm{A}(-\sqrt{3}a, 0)$ and $\mathrm{B}(0,-\sqrt{2}b)$, where a and b are real parameters and $ab \neq 0$. Then the locus of the centroid of $\triangle OAB$ is a circle of radius

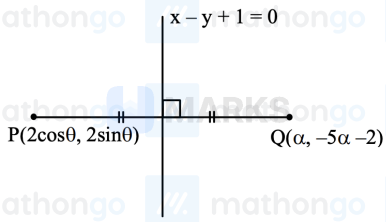
(1) $\frac{\frac{7}{3}}{3}$ (2) $\frac{8}{3}$ (3) $\frac{11}{3}$ (4) $\frac{5}{3}$

ANSWERS AND SOLUTIONS

1. 2 **2.** (1) **3.** (4) **4.** (4) **5.** (1) **6.** (3) **7.** (1) **8.** (1)

9. (2)

1.



(2)

Mid point of PQ lies on $x - y + 1 = 0$

$$\frac{2\cos\theta + \alpha}{2} - \frac{2\sin\theta - 5\alpha - 2}{2} + 1 = 0$$

$$2\cos\theta + \alpha - 2\sin\theta + 5\alpha + 2 + 2 = 0$$

$$\cos\theta - \sin\theta + 3\alpha + 2 = 0 \ ...(1)$$

$\because$ Slope of PQ is -1

$$\frac{2\sin\theta + 5\alpha + 2}{2\cos\theta - \alpha} = -1$$

$$2\sin\theta + 5\alpha + 2 = -2\cos\theta + \alpha$$

$$\sin\theta + \cos\theta + 2\alpha + 1 = 0 \ ...(2)$$

eliminate α from (1) and (2)

$$\Rightarrow \cos\theta + 5\sin\theta = 1 \ , \theta \in [0, 2\pi]$$

$$\Rightarrow 5 \times 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\sin^2\frac{\theta}{2}$$

$$\therefore \sin\frac{\theta}{2} = 0 \Rightarrow \cos\theta = 1$$

or

$$\sin\frac{\theta}{2} = 5 \Rightarrow \cos\theta = -\frac{12}{13}$$

Sum of all possible values of abscissa of point P is

$$= 2 \times 1 + 2\left(\frac{-12}{13}\right) = \frac{2}{13}$$

$\therefore$ 13 times sum of all possible values of abscissa of point P is 2.

2. **(1)** Let point of intersection $R(h, k)$

$$m_{BR} = m_{BP} \Rightarrow \frac{k}{h+2} = \frac{2\sin\alpha}{2\cos\alpha + 2} \Rightarrow \frac{k}{h+2} = \tan\frac{\alpha}{2}$$

$$m_{AR} = m_{AQ} \Rightarrow \frac{k}{h-2} = \frac{2\sin\beta}{2\cos\beta - 2} = \frac{\sin\beta}{\cos\beta - 1} = -\cot\frac{\beta}{2}$$

$$\frac{\alpha}{2} - \frac{\beta}{2} = \frac{\pi}{4}$$

$$\tan\left(\frac{\alpha}{2}-\frac{\beta}{2}\right)=\tan\frac{\pi}{4}=1$$

$$\frac{\tan\frac{\alpha}{2}-\tan\frac{\beta}{2}}{1+\tan\frac{\alpha}{2}\tan\frac{\beta}{2}}=1$$

$$\frac{\frac{k}{h+2}+\frac{h-2}{k}}{1+\left(\frac{k}{h+2}\right)\left(\frac{2-h}{k}\right)}=1\Rightarrow\frac{k^2+h^2-4}{\frac{k(h+2)}{\frac{4}{h+2}}}=1$$

$$\frac{h^2+k^2-4}{4k}=1$$

$$x^2+y^2-4y-4=0$$

3. **(4)** Circle 1: center $C_1=(-1,-4)$, radius r.
Circle 2: $(x-2)^2+(y-1)^2=9$, center $C_2=(2,1)$, radius 3.
Distance $d=\sqrt{9+25}=\sqrt{34}$.
For two intersection points: $|r-3|<\sqrt{34}<r+3$.
$r+3>\sqrt{34}\Rightarrow r>\sqrt{34}-3$ and $|r-3|<\sqrt{34}\Rightarrow r<\sqrt{34}+3$.
$r\in(\sqrt{34}-3,\ \sqrt{34}+3)$, so $\alpha=\sqrt{34}-3$, $\beta=\sqrt{34}+3$.
$\alpha\beta=(\sqrt{34})^2-9=34-9=25$.

4. **(4)** The circle through intersection of the ellipses is $S_1+\lambda S_2=0$.
$(1+4\lambda)x^2+(2+2\lambda)y^2-(6+20\lambda)x-(12+12\lambda)y+(23+35\lambda)=0$.
For a circle: $1+4\lambda=2+2\lambda\Rightarrow\lambda=\frac{1}{2}$.
Circle: $3x^2+3y^2-16x-18y+\frac{81}{2}=0\Rightarrow x^2+y^2-\frac{16}{3}x-6y+\frac{27}{2}=0$.
Centre $(a,b)=\left(\frac{8}{3},3\right)$, $r^2=\frac{64}{9}+9-\frac{27}{2}=\frac{47}{18}$.
$ab+18r^2=\frac{8}{3}\times3+18\times\frac{47}{18}=8+47=55$.

5. **(1)** From $\vec{c}\times\vec{a}=\vec{a}\times\vec{d}$ where $\vec{a}=2\hat{i}+3\hat{j}+4\hat{k}$, we get $(\vec{c}+\vec{d})\times\vec{a}=0$.
So $\vec{c}+\vec{d}=t\vec{a}$. Given $|\vec{c}+\vec{d}|=\sqrt{29}$: $|t|\cdot\sqrt{29}=\sqrt{29}$, so $t=\pm1$.
$(\vec{c}+\vec{d})\cdot(-7\hat{i}+2\hat{j}+3\hat{k})=\pm(-14+6+12)=\pm4$.
$\lambda_1=4$, $\lambda_2=-4$.
For circle: Coeff of x^2 = Coeff of y^2: $K^2=3K-2\Rightarrow K=1,2$.
Coeff of $xy=0$: $K^2-5K+4=0\Rightarrow K=1,4$.
Common value: $K=1$.

6. **(3)** Circle: $(x-2)^2+(y-3)^2=16$, center O = (2, 3), radius $r=4$.
For tangents from external point P touching at A and B with $\angle AOB=\pi/3$:
In quadrilateral OAPB: $\angle APB=\pi-\pi/3=2\pi/3$, so $\angle OPA=\pi/3$.
In right triangle OAP: $\sin(\pi/3)=\frac{4}{OP}$.

$OP = \frac{8}{\sqrt{3}} = \frac{8\sqrt{3}}{3}$.

Locus: $(x-2)^2 + (y-3)^2 = \frac{64}{3}$.

$3(x^2+y^2) - 12x - 18y - 25 = 0$.

7. **(1)** Circle C_1 has radius 5 and chord on line $y = x$ through origin.

Setting center at $(5, 0)$ (in region $x \geq 0$), the chord endpoints are $(0, 0)$ and $(5, 5)$ by solving $(t-5)^2 + t^2 = 25$.

Circle C_2 has this chord as diameter, so center is at $(\frac{5}{2}, \frac{5}{2})$ with radius $\frac{5\sqrt{2}}{2}$.

The chord of C_2 through $(2, 3)$ that is farthest from center is perpendicular to the radial direction from center to $(2, 3)$.

The radial direction is $(2 - \frac{5}{2}, 3 - \frac{5}{2}) = (-\frac{1}{2}, \frac{1}{2})$ with normal direction $(1, -1)$.

The chord equation is $1(x-2) - 1(y-3) = 0$, giving $x - y + 1 = 0$.

In form $x + ay + b = 0$, we have $a = -1$ and $b = 1$, so $a - b = -1 - 1 = -2$.

8. **(1)** Solving $\left|\frac{z-6i}{z-2i}\right| = 1 \Rightarrow y = 4$ .....(1)

(where $z = x + iy$)

Now solving $\left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5}$

$\Rightarrow x^2 + y^2 - 25x + 4y + 104 = 0$ .....(2)

Solving (1) & (2) $\Rightarrow z = 17 + 4i$ & $8 + 4i$

$\Rightarrow \sum |z|^2 = (17)^2 + (4)^2 + (8)^2 + (4)^2 = 385$

9. **(2)** Circle of radius 4 passes through O and points A($-\sqrt{3a}$, 0), B(0, $-\sqrt{2b}$).

Let center be C(h, k) on circle: $h^2 + k^2 = 16$.

Since A, B lie on circle: $3a^2 + 2\sqrt{3a}h + h^2 + k^2 = 16$ gives $h = -\frac{\sqrt{3a}}{2}$

And $h^2 + 2b + 2\sqrt{2b}k + k^2 = 16$ gives $k = -\frac{\sqrt{2b}}{2}$

Centroid G = $\left(\frac{-\sqrt{3a}}{3}, \frac{-\sqrt{2b}}{3}\right)$. Setting $x = \frac{-\sqrt{3a}}{3}$ and $y = \frac{-\sqrt{2b}}{3}$:

$a = 3x^2$ and $b = \frac{9y^2}{2}$

From $h^2 + k^2 = 16$: $\frac{3a}{4} + \frac{2b}{4} = 16 \rightarrow 3a + 2b = 64$

Substituting: $9x^2 + 9y^2 = 64 \rightarrow x^2 + y^2 = \frac{64}{9}$

Radius = $\frac{8}{3}$